

NUMERICAL SUPPLEMENT: CONTROLLED DENSITIES AND PRESERVED EVIDENCE

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Every new model–method run is retained, followed by the complete R50 numerical supplement and reconstructed R48 manuscript. The final fraction intervals and full binary covers are independently readable from the proof files. The inherited material remains historical evidence, not a claim of new execution.

1. EVERY CONTROLLED-DENSITY RUN

Common optimizes all feasible shared moments; Uniform is the exact policy-class counterfactual with spatially uniform final regret. A worker timer includes serialization and independent replay but not interpreter startup. The soft construction budget is not an all-in budget. Model primitives, the moment geometry, all cover nodes, and individual process high-water memory are in the accompanying JSON. All 54 runs are shown, with no selection by outcome.

TABLE I: Complete R51 execution records

Case	Class	L	U	Rel. %	Seconds	Nodes	KiB
A-base-5101	Common	2.091109	2.092107	0.0477	1.286	1277	22.9
A-base-5101	Uniform	2.105712	2.106684	0.0461	0.226	211	4.5
A-base-5102	Common	2.143995	2.144994	0.0465	1.345	1361	24.3
A-base-5102	Uniform	2.157516	2.158484	0.0448	0.216	197	4.2
A-base-5103	Common	2.076101	2.077102	0.0481	1.472	1489	26.5
A-base-5103	Uniform	2.089047	2.090022	0.0466	0.218	197	4.2
A-beta05-5101	Common	1.439977	1.440975	0.0692	1.153	1225	21.7
A-beta05-5101	Uniform	1.447427	1.448393	0.0666	0.207	187	4.0
A-beta05-5102	Common	1.473047	1.474046	0.0677	1.283	1315	23.3
A-beta05-5102	Uniform	1.479963	1.480927	0.0651	0.196	175	3.8
A-beta05-5103	Common	1.433896	1.434893	0.0695	1.308	1381	24.4
A-beta05-5103	Uniform	1.441019	1.441986	0.0670	0.199	181	3.9
A-beta09-5101	Common	2.482521	2.483520	0.0402	1.297	1305	23.6
A-beta09-5101	Uniform	2.500542	2.501530	0.0395	0.219	203	4.4
A-beta09-5102	Common	2.547193	2.548193	0.0392	1.382	1399	25.1
A-beta09-5102	Uniform	2.563927	2.564916	0.0385	0.207	189	4.1
A-beta09-5103	Common	2.461859	2.462858	0.0405	1.431	1471	26.3
A-beta09-5103	Uniform	2.477763	2.478757	0.0400	0.219	195	4.2
A-eps01-5101	Common	2.310251	2.311250	0.0432	1.009	1151	20.8
A-eps01-5101	Uniform	2.312171	2.313140	0.0418	0.070	85	2.0
A-eps01-5102	Common	2.365862	2.366859	0.0421	0.883	987	17.9
A-eps01-5102	Uniform	2.366880	2.367875	0.0420	0.052	63	1.6
A-eps01-5103	Common	2.299431	2.300429	0.0433	0.882	977	17.7
A-eps01-5103	Uniform	2.300156	2.301145	0.0429	0.053	63	1.6
A-eps03-5101	Common	1.891408	1.892407	0.0528	1.485	1485	26.7
A-eps03-5101	Uniform	1.905751	1.906684	0.0489	0.207	193	4.2
A-eps03-5102	Common	1.944203	1.945203	0.0514	1.524	1557	27.8
A-eps03-5102	Uniform	1.957564	1.958484	0.0469	0.215	181	4.0
A-eps03-5103	Common	1.875696	1.876696	0.0532	1.671	1655	29.6
A-eps03-5103	Uniform	1.889071	1.890022	0.0503	0.204	183	4.1
A-no-control-5101	Common	2.134886	2.135873	0.0462	0.164	185	4.1
A-no-control-5101	Uniform	2.134886	2.135873	0.0462	0.154	185	4.1

Case	Class	L	U	Rel. %	Seconds	Nodes	KiB
A-no-control-5102	Common	2.193860	2.194860	0.0455	0.314	345	7.0
A-no-control-5102	Uniform	2.193860	2.194860	0.0455	0.325	345	7.0
A-no-control-5103	Common	2.124132	2.125120	0.0465	0.282	293	6.1
A-no-control-5103	Uniform	2.124132	2.125120	0.0465	0.291	293	6.1
A-reverse-5101	Common	2.147824	2.148822	0.0464	0.516	561	10.4
A-reverse-5101	Uniform	2.147838	2.148822	0.0457	0.055	59	1.6
A-reverse-5102	Common	2.209619	2.210619	0.0452	0.783	843	15.2
A-reverse-5102	Uniform	2.209738	2.210619	0.0398	0.072	77	2.0
A-reverse-5103	Common	2.141464	2.142460	0.0464	1.094	1207	21.5
A-reverse-5103	Uniform	2.141487	2.142460	0.0454	0.084	89	2.2
B-base-5101	Common	1.553413	1.554414	0.0643	3.326	3453	61.3
B-base-5101	Uniform	1.553619	1.554545	0.0595	0.098	123	2.9
B-base-5102	Common	1.566057	1.567057	0.0638	4.743	5031	89.6
B-base-5102	Uniform	1.566791	1.567740	0.0605	0.129	161	3.6
B-base-5103	Common	1.449564	1.450564	0.0689	1.386	1449	26.0
B-base-5103	Uniform	1.449585	1.450564	0.0674	0.069	83	2.1
C-base-5101	Common	2.468025	2.469025	0.0405	1.665	1827	33.5
C-base-5101	Uniform	2.469444	2.470424	0.0396	0.049	59	1.6
C-base-5102	Common	2.502504	2.503503	0.0399	1.809	1913	35.1
C-base-5102	Uniform	2.504579	2.505574	0.0397	0.055	65	1.7
C-base-5103	Common	2.518343	2.519343	0.0397	1.924	2129	39.0
C-base-5103	Uniform	2.524629	2.525583	0.0377	0.083	99	2.4

2. INSPECTION AND ARITHMETIC BOUNDARY

The standalone reader imports no constructor, numerical solver, or project mathematical library. It enumerates all moment corners afresh, integrates a separately derived binding-mixture expression, and uses a different logarithm truncation and precision. It checks raw-model conditions, feasible common moments, every binary partition, every discarded rectangle, all inherited lower bounds, the final upper cost, and target status. Eight contract tests include tampered lower bounds, invalid moments, an omitted cover child, false moment-set pruning, high-precision logarithm and integral cross-checks, and an independent SciPy row-LP comparison. These are code-path and arithmetic checks, not independent scientific replication.

A fresh reader can start from `BENCHMARK.md` and one proof without importing the constructor. `WORK_CURVES.json` retains all available power-of-two-node checkpoints with elapsed construction time. Their bounds do not include hypothetical future verification costs. Final all-in cost is separately reported. The original proof files retain exact fractions; printed decimals are not the acceptance boundary.

3. FULL INHERITED SUPPLEMENT

The following pages reproduce `SUPP_R50.pdf` from the pinned R50 parent. They include all 78 earlier reset runs, finite failures, price audits, earlier responses to measurement concerns, and the reconstructed complete R48 manuscript. No inherited result is counted as a new controlled-density run.

SUPPLEMENT TO CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

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This supplement retains every R50 model–method result, exact-proof reference, scale-free decision measure, finite diagnostic, and price-cap audit. It also includes a reconstructed copy of the complete R48 article with its missing tables restored from the original records. The historical text is not a description of the current revision’s evidence.

1. READING THE NUMERICAL EVIDENCE

The ordinary entry points are `ECTA_R50.tex`, `SUPP_R50.tex`, and `RESPONSE_R50.tex`. The root index `R50_REVIEW.md` identifies the reviewed commit and the immutable source freeze. Exact numerical decisions use rational values in the JSON records; tabulated decimal endpoints are outward rounded in the current tables. Widths and other descriptive quantities are rounded for display, not used for acceptance tests.

The frozen primary study has 39 diffuse models and two backends. Each retained grid proof is checked, not only the final successful level. The separate finite study reruns all twelve historical models at two soft construction budgets with two complete constructors. A started exact operation is not interrupted. The scientific record separates construction, serialization, Python verification, and total measured worker cost. JavaScript checks and contract tests are additional costs recorded separately. High-water memory is process-specific; Python interpreter launch before the worker timer is excluded from worker time. No table equates a declared construction budget to exact elapsed time.

2. DIFFUSE-LAW MODELS: EVERY BACKEND AND ENDPOINT

DS denotes dual simplex; IPM denotes interior point. N is the final grid. Exact cost endpoints, policy probabilities, cumulative moments, multiplier fields, numerical LP statuses, source-model identity, proof hashes, and all intermediate grid levels reside under `revisions/2026-09-26-r50`. A positive constant rescaling of implementation costs requires rescaling the absolute accuracy by the same constant.

TABLE I: All 78 diffuse model–backend results

Case	Method	N	L	U	Width	Seconds	KiB
T16-401	ds	256	3.714351	3.714610	0.000258	4.681	25.0
T16-401	ipm	256	3.714351	3.714610	0.000258	5.560	25.0
T16-503	ds	256	3.843817	3.844282	0.000464	4.645	25.1
T16-503	ipm	256	3.843817	3.844282	0.000464	6.218	25.1
T16-607	ds	64	4.174783	4.175660	0.000876	1.378	3.8
T16-607	ipm	64	4.174784	4.175660	0.000876	1.428	3.8
T32-401	ds	256	4.426358	4.426617	0.000258	8.874	25.9
T32-401	ipm	256	4.426358	4.426617	0.000258	9.900	25.9
T32-503	ds	256	4.582065	4.582531	0.000466	9.150	28.2
T32-503	ipm	256	4.582065	4.582531	0.000466	9.764	28.2
T32-607	ds	64	4.885056	4.885934	0.000877	2.376	4.6
T32-607	ipm	64	4.885056	4.885934	0.000876	2.461	4.6

Case	Method	N	L	U	Width	Seconds	KiB
T4-401	ds	64	1.457270	1.458094	0.000823	0.660	3.1
T4-401	ipm	64	1.457270	1.458094	0.000823	0.663	3.1
T4-503	ds	256	1.739387	1.739813	0.000425	1.471	15.1
T4-503	ipm	256	1.739387	1.739813	0.000425	1.539	15.1
T4-607	ds	64	1.962313	1.963107	0.000793	0.644	2.2
T4-607	ipm	64	1.962313	1.963107	0.000792	0.652	2.2
base-401	ds	64	2.542613	2.543585	0.000971	0.911	5.4
base-401	ipm	64	2.542613	2.543585	0.000971	0.937	5.4
base-503	ds	256	2.693783	2.694219	0.000435	2.486	17.0
base-503	ipm	256	2.693784	2.694219	0.000435	2.695	17.0
base-607	ds	64	3.086552	3.087348	0.000795	0.889	2.4
base-607	ipm	64	3.086558	3.087348	0.000789	0.888	2.4
beta50-401	ds	64	0.853997	0.854754	0.000756	0.924	5.3
beta50-401	ipm	64	0.853997	0.854754	0.000756	0.931	5.3
beta50-503	ds	256	1.001130	1.001525	0.000394	2.476	17.1
beta50-503	ipm	256	1.001130	1.001525	0.000394	2.801	17.1
beta50-607	ds	64	1.155028	1.155819	0.00079	0.886	2.3
beta50-607	ipm	64	1.155029	1.155819	0.000789	0.886	2.3
beta97-401	ds	256	3.258958	3.259229	0.00027	2.622	17.5
beta97-401	ipm	256	3.258958	3.259229	0.00027	2.998	17.5
beta97-503	ds	256	3.345526	3.345975	0.000447	2.556	16.9
beta97-503	ipm	256	3.345526	3.345975	0.000447	2.717	16.9
beta97-607	ds	64	3.865074	3.865873	0.000798	0.883	2.4
beta97-607	ipm	64	3.865083	3.865873	0.000789	0.898	2.4
beta99-401	ds	256	3.505347	3.505629	0.000281	2.660	17.3
beta99-401	ipm	256	3.505347	3.505629	0.000281	3.113	17.3
beta99-503	ds	256	3.565329	3.565781	0.000451	2.532	16.7
beta99-503	ipm	256	3.565329	3.565781	0.000451	2.767	16.7
beta99-607	ds	64	4.130682	4.131483	0.0008	0.888	2.4
beta99-607	ipm	64	4.130693	4.131483	0.000789	0.904	2.4
eps02-401	ds	64	2.559433	2.560405	0.000971	0.908	5.2
eps02-401	ipm	64	2.559433	2.560405	0.000971	0.935	5.3
eps02-503	ds	256	2.710995	2.711431	0.000435	2.499	16.8
eps02-503	ipm	256	2.710995	2.711431	0.000435	2.698	16.8
eps02-607	ds	64	3.112137	3.112937	0.000799	0.883	2.4
eps02-607	ipm	64	3.112147	3.112937	0.000789	0.902	2.4
eps15-401	ds	64	2.486545	2.487517	0.000971	0.907	5.4
eps15-401	ipm	64	2.486545	2.487517	0.000971	0.931	5.5
eps15-503	ds	256	2.636411	2.636846	0.000435	2.492	17.4
eps15-503	ipm	256	2.636411	2.636846	0.000435	2.687	17.4
eps15-607	ds	64	3.001256	3.002053	0.000795	0.875	2.4
eps15-607	ipm	64	3.001263	3.002053	0.000789	0.893	2.4
m2-401	ds	256	2.634198	2.634588	0.000389	1.874	19.5
m2-401	ipm	256	2.634198	2.634588	0.000389	1.918	19.5
m2-503	ds	256	3.032092	3.032605	0.000512	1.780	8.2
m2-503	ipm	256	3.032092	3.032605	0.000512	1.840	8.2
m2-607	ds	256	2.498228	2.498536	0.000307	1.929	23.2
m2-607	ipm	256	2.498228	2.498536	0.000307	2.021	23.2
m5-401	ds	64	2.661519	2.662385	0.000866	1.183	2.6
m5-401	ipm	64	2.661522	2.662385	0.000862	1.202	2.6
m5-503	ds	256	3.031576	3.031879	0.000302	3.929	13.3
m5-503	ipm	256	3.031576	3.031879	0.000302	4.398	13.3
m5-607	ds	256	2.583105	2.583389	0.000282	4.016	21.1
m5-607	ipm	256	2.583105	2.583389	0.000282	4.134	21.1
slope0-401	ds	64	2.534421	2.535320	0.000898	0.898	2.6
slope0-401	ipm	64	2.534421	2.535320	0.000898	0.927	2.6
slope0-503	ds	256	2.684960	2.685220	0.000259	2.489	5.7

Case	Method	N	L	U	Width	Seconds	KiB
slope0-503	ipm	256	2.684960	2.685220	0.000259	2.679	6.0
slope0-607	ds	64	3.081307	3.081937	0.000629	0.891	1.4
slope0-607	ipm	64	3.081316	3.081937	0.00062	0.888	1.5
slope80-401	ds	256	2.547613	2.547956	0.000343	2.566	17.3
slope80-401	ipm	256	2.547613	2.547956	0.000343	3.214	17.3
slope80-503	ds	256	2.697757	2.698491	0.000734	2.560	21.4
slope80-503	ipm	256	2.697757	2.698491	0.000733	2.884	21.4
slope80-607	ds	256	3.090822	3.091130	0.000308	2.457	7.2
slope80-607	ipm	256	3.090822	3.091130	0.000307	2.539	7.2

3. SCALE-FREE DECISION ACCURACY

C^{op} is the exact implementation expense of an operating-best policy. The displayed saving is $C^{\text{op}} - U$, a guaranteed feasible cost improvement. The relative width divides by U ; normalized decision loss divides by C^{op} . These quantities do not assert market calibration, empirical maintenance savings, or a welfare estimate. Only one backend is listed here because the model and benchmark expense are the same for both; the complete JSON provides both results.

TABLE II: Decision accuracy in all 39 diffuse models, dual-simplex proposals

Case	C^{op}	Saving lower	Relative width %	Norm. loss %
T16-401	3.745769	0.031159	0.0069	0.0069
T16-503	3.877146	0.032864	0.0121	0.0120
T16-607	4.219309	0.043649	0.0210	0.0208
T32-401	4.457776	0.031159	0.0058	0.0058
T32-503	4.615433	0.032902	0.0102	0.0101
T32-607	4.929582	0.043649	0.0179	0.0178
T4-401	1.484048	0.025955	0.0564	0.0554
T4-503	1.768373	0.028560	0.0244	0.0240
T4-607	2.005754	0.042648	0.0404	0.0395
base-401	2.571618	0.028034	0.0382	0.0378
base-503	2.722905	0.028686	0.0161	0.0160
base-607	3.129996	0.042648	0.0257	0.0254
beta50-401	0.880457	0.025704	0.0885	0.0859
beta50-503	1.027452	0.025927	0.0394	0.0384
beta50-607	1.198466	0.042648	0.0684	0.0659
beta97-401	3.288765	0.029537	0.0083	0.0082
beta97-503	3.375492	0.029518	0.0134	0.0133
beta97-607	3.908520	0.042648	0.0206	0.0204
beta99-401	3.535715	0.030087	0.0080	0.0079
beta99-503	3.595559	0.029779	0.0127	0.0126
beta99-607	4.174130	0.042648	0.0194	0.0192
eps02-401	2.571618	0.011213	0.0379	0.0378
eps02-503	2.722905	0.011475	0.0160	0.0160
eps02-607	3.129996	0.017059	0.0257	0.0255
eps15-401	2.571618	0.084101	0.0390	0.0378
eps15-503	2.722905	0.086059	0.0165	0.0160
eps15-607	3.129996	0.127944	0.0265	0.0254
m2-401	2.662664	0.028076	0.0148	0.0146
m2-503	3.065770	0.033165	0.0169	0.0167
m2-607	2.523056	0.024521	0.0123	0.0122
m5-401	2.734664	0.072279	0.0325	0.0317
m5-503	3.104689	0.072810	0.0100	0.0097

Case	C^{op}	Saving lower	Relative width %	Norm. loss %
m5-607	2.631093	0.047705	0.0109	0.0107
slope0-401	2.571618	0.036298	0.0354	0.0349
slope0-503	2.722905	0.037686	0.0096	0.0095
slope0-607	3.129996	0.048059	0.0204	0.0201
slope80-401	2.571618	0.023662	0.0134	0.0133
slope80-503	2.722905	0.024414	0.0272	0.0269
slope80-607	3.129996	0.038866	0.0099	0.0098

4. ALL FINITE CONSTRUCTIONS, INCLUDING UNFINISHED RUNS

These are retrospective tests on the R48 raw models, not additional source-frozen environments. Their price constructor uses the original probability-space cover; the budget constructor uses the new shared-regret coordinates. Complete-constructor differences include local-proposal allocations and branch rules. The table therefore does not attribute a timing difference causally to the coordinate change alone.

TABLE III: All 48 finite runs

Case	Method	Cap	L	U	Width	Seconds	Nodes	MiB
inventory0	budget	2	1.976949	1.982232	0.00528	3.24	307	78.1
inventory0	budget	8	1.978951	1.982232	0.00328	5.01	511	78.5
inventory0	price	2	1.981340	1.982232	0.000892	0.90	9	80.2
inventory0	price	8	1.981340	1.982232	0.000892	0.92	9	80.4
inventory1	budget	2	4.966182	5.044754	0.0786	3.16	55	77.8
inventory1	budget	8	4.977586	5.044754	0.0672	11.56	255	80.6
inventory1	price	2	5.019833	5.044754	0.0249	3.62	9	99.8
inventory1	price	8	5.023170	5.044754	0.0216	12.91	37	116.3
inventory2	budget	2	10.196728	10.246152	0.0494	3.36	9	83.0
inventory2	budget	8	10.196728	10.244317	0.0476	11.78	45	83.9
inventory2	price	2	10.196728	10.248454	0.0517	3.69	1	115.9
inventory2	price	8	10.227356	10.244317	0.017	14.43	5	153.8
queue0	budget	2	0.230545	0.230546	9.3e-09	0.47	1	75.9
queue0	budget	8	0.230545	0.230546	9.3e-09	0.46	1	75.8
queue0	price	2	0.230545	0.230546	9.3e-09	0.52	1	77.4
queue0	price	8	0.230545	0.230546	9.3e-09	0.50	1	77.4
queue1	budget	2	7.388825	7.424602	0.0358	3.19	57	78.0
queue1	budget	8	7.388924	7.424602	0.0357	11.58	241	80.4
queue1	price	2	7.419435	7.424602	0.00517	3.68	11	101.1
queue1	price	8	7.421587	7.424602	0.00301	13.73	41	141.5
queue2	budget	2	10.617117	10.652032	0.0349	3.23	9	83.0
queue2	budget	8	10.617117	10.652032	0.0349	11.81	47	83.5
queue2	price	2	10.636976	10.652032	0.0151	3.93	1	120.2
queue2	price	8	10.638018	10.652032	0.014	15.45	7	166.9
ties0	budget	2	1.067956	1.074850	0.00689	3.18	309	77.6
ties0	budget	8	1.069417	1.074850	0.00543	5.10	511	78.5
ties0	price	2	1.057376	1.074850	0.0175	3.71	61	93.4
ties0	price	8	1.065757	1.074850	0.00909	13.51	235	119.5
ties1	budget	2	0.000000	0.000000	0	0.59	1	77.0
ties1	budget	8	0.000000	0.000000	0	0.60	1	77.1
ties1	price	2	0.000000	0.000000	0	0.88	1	86.0
ties1	price	8	0.000000	0.000000	0	0.88	1	86.5
ties2	budget	2	6.082702	6.135198	0.0525	3.15	11	83.0
ties2	budget	8	6.082702	6.135181	0.0525	11.30	59	84.0

Case	Method	Cap	L	U	Width	Seconds	Nodes	MiB
ties2	price	2	6.082702	6.139907	0.0572	5.06	3	145.6
ties2	price	8	6.082702	6.135106	0.0524	13.01	7	176.4
warranty0	budget	2	3.107625	3.110700	0.00307	3.18	309	77.4
warranty0	budget	8	3.107625	3.110700	0.00307	5.00	511	78.2
warranty0	price	2	3.109712	3.110700	0.000987	0.71	5	78.2
warranty0	price	8	3.109712	3.110700	0.000987	0.71	5	78.9
warranty1	budget	2	8.205716	8.236030	0.0303	3.14	55	77.8
warranty1	budget	8	8.205716	8.236030	0.0303	11.55	245	80.2
warranty1	price	2	8.220455	8.236030	0.0156	3.62	9	102.4
warranty1	price	8	8.222081	8.236030	0.0139	13.26	39	137.8
warranty2	budget	2	8.897888	8.926491	0.0286	3.13	9	82.9
warranty2	budget	8	8.897888	8.926491	0.0286	12.01	53	83.8
warranty2	price	2	8.906663	8.926491	0.0198	3.68	1	128.6
warranty2	price	8	8.910482	8.926491	0.016	15.06	7	165.8

4.1. Endpoint movement and witness accuracy

U_0 is the initial upper policy, L_0 is the root lower endpoint, $\Delta U = U_0 - U$, and $\Delta L = L - L_0$. The guard ratio divides the maximum operating-witness width by $(1 - \beta)\varepsilon$. A small ratio rules out that particular enclosure error as the sole cause of a large optimization interval. It does not validate an efficient operating oracle in a harder economy. B and P denote the budget and probability-price constructors.

TABLE IV: Root-to-final movement and operating witnesses

Case	M	Cap	U_0	L_0	ΔU	ΔL	Bits	Guard ratio
inventory0	B	2	1.98859	1.97012	0.0064	0.0068	32	3e-07
inventory0	B	8	1.98859	1.97012	0.0064	0.0088	32	3e-07
inventory0	P	2	1.98859	1.98122	0.0064	0.00012	32	3e-07
inventory0	P	8	1.98859	1.98122	0.0064	0.00012	32	3e-07
inventory1	B	2	5.23422	4.96024	0.19	0.0059	40	9e-09
inventory1	B	8	5.23422	4.96024	0.19	0.017	40	9e-09
inventory1	P	2	5.23422	4.99789	0.19	0.022	40	9e-09
inventory1	P	8	5.23422	4.99789	0.19	0.025	40	9e-09
inventory2	B	2	10.53513	10.19673	0.29	3.2e-17	48	5e-10
inventory2	B	8	10.53513	10.19673	0.29	9.7e-17	48	5e-10
inventory2	P	2	10.53513	10.19673	0.29	0	48	5e-10
inventory2	P	8	10.53513	10.22736	0.29	1.2e-10	48	5e-10
queue0	B	2	0.23055	0.23055	0	0	32	4e-07
queue0	B	8	0.23055	0.23055	0	0	32	4e-07
queue0	P	2	0.23055	0.23055	0	0	32	4e-07
queue0	P	8	0.23055	0.23055	0	0	32	4e-07
queue1	B	2	7.45534	7.38877	0.031	5.3e-05	40	9e-09
queue1	B	8	7.45534	7.38877	0.031	0.00015	40	9e-09
queue1	P	2	7.45534	7.41764	0.031	0.0018	40	9e-09
queue1	P	8	7.45534	7.41764	0.031	0.0039	40	9e-09
queue2	B	2	10.72600	10.61712	0.074	0	48	5e-10
queue2	B	8	10.72600	10.61712	0.074	0	48	5e-10
queue2	P	2	10.72600	10.63698	0.074	0	48	5e-10
queue2	P	8	10.72600	10.63698	0.074	0.001	48	5e-10
ties0	B	2	1.07485	1.05897	0	0.009	8	0
ties0	B	8	1.07485	1.05897	0	0.01	8	0
ties0	P	2	1.07485	1.05568	0	0.0017	8	0

Case	M	Cap	U_0	L_0	ΔU	ΔL	Bits	Guard ratio
ties0	P	8	1.07485	1.05568	0	0.01	8	0
ties1	B	2	0.00000	0.00000	0	0	8	0
ties1	B	8	0.00000	0.00000	0	0	8	0
ties1	P	2	0.00000	0.00000	0	0	8	0
ties1	P	8	0.00000	0.00000	0	0	8	0
ties2	B	2	6.32506	6.08270	0.19	0	8	0
ties2	B	8	6.32506	6.08270	0.19	0	8	0
ties2	P	2	6.32506	6.08270	0.19	0	8	0
ties2	P	8	6.32506	6.08270	0.19	0	8	0
warranty0	B	2	3.11457	3.10763	0.0039	0	32	3e-07
warranty0	B	8	3.11457	3.10763	0.0039	0	32	3e-07
warranty0	P	2	3.11457	3.10893	0.0039	0.00078	32	3e-07
warranty0	P	8	3.11457	3.10893	0.0039	0.00078	32	3e-07
warranty1	B	2	8.41498	8.20572	0.18	7.1e-16	40	1e-08
warranty1	B	8	8.41498	8.20572	0.18	7.1e-16	40	1e-08
warranty1	P	2	8.41498	8.21218	0.18	0.0083	40	1e-08
warranty1	P	8	8.41498	8.21218	0.18	0.0099	40	1e-08
warranty2	B	2	8.96030	8.89789	0.034	0	48	5e-10
warranty2	B	8	8.96030	8.89789	0.034	0	48	5e-10
warranty2	P	2	8.96030	8.90666	0.034	0	48	5e-10
warranty2	P	8	8.96030	8.90666	0.034	0.0038	48	5e-10

4.2. Proof resources and rational size

The variable column reports the maximum numerical node-variable count recorded by the probability-price tree. Zero for the budget method means its row programs are solved by exact action/pair enumeration, not that the policy has no variables. Leaves include retained and terminal nodes. Numerator and denominator columns are maximum bit lengths found by a complete recursive scan of the proof. Time curves, LP statuses, mask attempts, local-proposal status, operating-oracle time, and price-amplification budgets are recorded individually in `results/diagnostics.json` and `results/work_curves.csv`.

TABLE V: Numerical and certificate resources

Case	M	Cap	Vars	Leaves	KiB	Check sec.	Num. bits	Den. bits
inventory0	B	2	0	154	23.8	0.792	231	230
inventory0	B	8	0	256	39.3	1.302	231	230
inventory0	P	2	222	5	9.6	0.176	851	851
inventory0	P	8	222	5	9.6	0.178	851	851
inventory1	B	2	0	28	12.9	0.731	408	405
inventory1	B	8	0	128	35.1	3.116	475	473
inventory1	P	2	998	5	57.6	1.071	3406	3411
inventory1	P	8	998	19	218.9	4.309	3408	3413
inventory2	B	2	0	5	21.7	0.650	733	729
inventory2	B	8	0	23	28.4	3.069	733	729
inventory2	P	2	4008	1	27.1	0.967	6673	6670
inventory2	P	8	4008	3	119.3	4.401	13808	13814
queue0	B	2	0	1	2.1	0.006	80	82
queue0	B	8	0	1	2.1	0.006	80	82
queue0	P	2	222	1	2.7	0.029	189	191
queue0	P	8	222	1	2.7	0.030	189	191
queue1	B	2	0	29	18.5	0.742	520	517
queue1	B	8	0	121	53.2	3.106	540	537

Case	M	Cap	Vars	Leaves	KiB	Check sec.	Num. bits	Den. bits
queue1	P	2	998	6	61.8	1.132	3507	3514
queue1	P	8	998	21	228.7	4.890	3513	3521
queue2	B	2	0	5	20.2	0.693	689	685
queue2	B	8	0	24	21.7	3.314	689	685
queue2	P	2	4008	1	45.4	1.076	13457	13463
queue2	P	8	4008	4	139.0	5.724	13483	13489
ties0	B	2	0	155	12.6	0.755	132	132
ties0	B	8	0	256	19.4	1.288	135	135
ties0	P	2	222	31	14.3	1.224	135	135
ties0	P	8	222	118	93.3	4.941	137	137
ties1	B	2	0	1	2.4	0.021	32	57
ties1	B	8	0	1	2.4	0.021	32	57
ties1	P	2	1250	1	2.7	0.130	32	57
ties1	P	8	1250	1	2.7	0.131	32	57
ties2	B	2	0	6	10.6	0.557	412	409
ties2	B	8	0	30	12.3	2.714	404	401
ties2	P	2	5384	2	12.4	1.662	415	413
ties2	P	8	5384	4	15.2	3.953	404	401
warranty0	B	2	0	155	14.2	0.774	210	208
warranty0	B	8	0	256	22.9	1.289	210	208
warranty0	P	2	222	3	5.6	0.100	820	819
warranty0	P	8	222	3	5.6	0.100	820	819
warranty1	B	2	0	28	9.2	0.722	433	430
warranty1	B	8	0	123	15.5	3.122	433	430
warranty1	P	2	1124	5	33.8	1.127	3235	3241
warranty1	P	8	1124	20	122.5	4.663	3256	3262
warranty2	B	2	0	5	17.8	0.611	663	660
warranty2	B	8	0	27	19.7	3.300	663	660
warranty2	P	2	4800	1	37.8	0.974	10489	10494
warranty2	P	8	4800	4	103.3	5.355	10263	10268

5. THE PRICE-FAMILY GAP, RATHER THAN THE COMMON-POLICY GAP

For every historical unfinished row, all positive-support masks are either supplied with a feasible occupation upper witness or pruned using the masked cost of an explicitly checked feasible policy. The global upper bound covers the uncapped ideal clipped price optimum. Subtracting the capped replay value yields the displayed loss upper bound. It is not an upper bound on the gap to the original common-policy optimum. Small support, not the total number of states, determines the number of masks in the large R48 models.

TABLE VI: Uncapped-upper minus capped replay: certified price-loss bounds

Case	Cap 256	Cap 1024	Cap 4096	Cap 16384	Audit sec.
inventory1	7.93e-15	7.93e-15	7.93e-15	7.93e-15	2.20
inventory2	1.02e-13	1.02e-13	1.02e-13	1.03e-13	1.65
queue1	1.24e-13	1.24e-13	1.24e-13	1.24e-13	2.30
queue2	0.109	0.109	0.109	0.109	1.67
ties0	6.81e-15	6.81e-15	6.81e-15	6.81e-15	0.19
ties2	2.92e-09	2.92e-09	2.92e-09	2.92e-09	1.72
warranty1	0.209	0.209	0.209	0.209	2.29
warranty2	5.8e-08	7.82e-08	4.71e-12	4.71e-12	1.65

5.1. Shared-successor compatibility and rational conditioning

For each nonzero incoming flow f_{tij} of an occupation witness, the implied successor allocation is α_{tij}/f_{tij} . The diagnostic records the largest spread of that quantity across incoming edges to one successor, maximized over retained occupation witnesses. A positive spread directly witnesses a lack of shared successor allocation in that particular feasible relaxation point. It is not a lower bound on the optimal common-policy gap, nor a regression claiming that spread causes observed tree time. Edges with zero flow have both allocation masses zero and impose no ratio restriction. Price and rational-size summaries distinguish numerical conditioning from arithmetic validity.

TABLE VII: Occupation-compatibility diagnostics

Case	Max budget spread	Max price	Num. bits	Den. bits
inventory1	0.01	117	1005	1003
inventory2	0.005	199	1946	1989
queue1	0.01	74.2	1360	1402
queue2	1.61e-06	195	2375	2373
ties0	0.02	58	253	294
ties2	0.005	164	1196	1194
warranty1	0.00873	63.3	1009	1006
warranty2	0.005	74.5	1876	1904

6. MINIMAL EXTERNAL-FACING BENCHMARK SPECIFICATION

A finite instance is the tuple $(n, T, m, \beta, \varepsilon, \nu, r, k, P, g)$ with rational entries, stochastic transition rows, and nonnegative implementation costs. A candidate policy is one simplex row for each (t, i) . Evaluate V , J^p , and C^p backward using the recursions in the article, require $V_{ti} - J_{ti}^p \leq \varepsilon$ for every state and date, and minimize νC_0^p . No hidden initial plan lottery is allowed. Third parties can reconstruct this optimization target from the JSON without importing project code.

A reset instance instead specifies T, m, β, ε , action productivities γ_a , affine coefficients of g_t and k_{ta} , a uniform reset kernel, and a uniform initial law. A deployed cell policy is checked at both endpoints of every cell. The continuation moment is the full cumulative quantity $M_t = \int p_t d_t + \beta M_{t+1}$, not just a one-period average. Its lower certificate minimizes over all Borel policy rows before integration. The Python and JavaScript readers implement these formulas separately. Both remain within-project implementations.

7. HISTORICAL R48 ARTICLE: PRESERVATION NOTICE

The following pages reproduce the complete modular R48 article associated with reviewed head 77b90a92250694271265e91f9624b7d6aacb0dea. Only input paths are redirected to the archived source directory and the two missing tables are reconstructed from the original recorded endpoints. The R48 narrative, including its former numerical boundaries and illustrative monetary conversion, remains historical. It must not be read as the R50 article's current empirical claims. All earlier repository revisions remain unchanged.

CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

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We study minimum-cost revision of an installed policy with operating protection after every state–date restart and one common randomized Markov continuation. We characterize the supremum of restart-price certificates by an occupation-flow linear program and identify the shared successor-budget equations that restore the original constrained problem. A strict separation example distinguishes the price relaxation from both a constant-multiplier bound and the constrained optimum. An end-to-end finite algorithm constructs operating enclosures, feasible policies, lower certificates, and covering refinements without access to exact optimal operating values or inherited incumbents. Its accuracy rule adapts arithmetic precision. For controlled finite-atomic transitions and a finitely supported initial law, an exact restriction–extension theorem transfers paired finite certificates to the full Borel-state problem. A source-frozen evaluation produces four target hits in twelve new primary environments for each of two independently checked formulations; the price formulation tightens all eight unclosed intervals. Two positive-cost observed-fleet instances of the original affine dynamics have paired continuum certificates. These finite-initial-law results do not close the thirty earlier positive-cost intervals under the uniform initial law. All runs, costs, and remaining intervals are reported.

KEYWORDS: Dynamic programming, Policy revision, Common Markov policies, Constrained control, Verified computation.

1. INTRODUCTION

An organization may promise operating protection whenever a decision rule is restarted, while evaluating the expense of implementing that rule under its actual initial population. These are different restrictions on a single policy. The first is a statewise performance contract; the second is an ex ante cost objective. Actions alter future states and therefore alter future implementation expense. A collection of inexpensive plans, each optimized for a different restart, need not define one policy at their common successor states.

We retain this distinction throughout. The decision variable is a Borel randomized Markov policy with one action distribution at each state and date. Its randomization is fresh conditional on that state and date; a hidden initial lottery over persistent plans is not substituted for the common continuation. The target is the least implementation cost subject to every restart’s operating-value shortfall being at most a prescribed allowance. Implementation cost can charge departures from an installed rule and can include an administratively specified realized-action charge. A fixed organizational cost of administering lotteries is a separate contract term.

The central result is a theorem–algorithm connection. First, we identify exactly what the restart-price family optimizes. Its supremum is the dual of an occupation-flow relaxation in which an edge may allocate the successor regret budget differently from other edges arriving at the same state. The correction in the price Bellman recursion is therefore not a free-standing strong-duality theorem for the original constrained problem. Adding common successor-budget equations makes the occupation formulation exact. A rational two-period example has a constant-multiplier value of $5/4$, a restart-price supremum of $27/16$, and an original common-policy optimum of $7/4$. The equalities locate both the gain from restart prices and the coupling which the global algorithm must restore.

Second, we construct both sides of the interval from primitive inputs. Directed Bellman enclosures are computed before any price or policy is proposed. A backward conditional-cost minimization supplies an initial feasible policy; a local numerical proposal is repaired against its actual continuation and the upper operating enclosure. Price proposals use only the lower and upper operating enclosures. Two different bilinear formulations use one common set of state–date probabilities. An independent rational reader reconstructs the relevant inequalities and the entire covering tree. The stopping theorem accounts for the enclosure width, policy-box diameter, and deployment quantization. Exact operating ties create no inverse-gap singularity in this argument. The sufficient tree bound is exponential, and the experiment reports rather than conceals its consequences.

Third, a restriction–extension theorem gives a matched controlled-continuum result for a finitely observed initial population. With finitely many transition atoms per action, the set of states reached from that population under *all* action and shock sequences is finite at a finite horizon. Restriction to that set preserves the operating problem. More importantly, every feasible common policy on the set extends to a feasible Borel policy on the entire state space by choosing an operating-best action against the already extended continuation off the set. This proves equality of the finite and continuum revision optima, not merely validity of one endpoint. The dynamics may move the continuous state in an action-dependent and noninvertible manner. No autonomous type label or transition smoothing is required.

This result has an exact boundary. A finite observed initial law is the objective law, not an empirical approximation to an unobserved uniform population. Atomic laws need not approximate a continuous law in the topology required for this all-restart constrained optimization. Accordingly, the two observed-fleet calculations below do not replace the original uniform-initial-law cohort. The thirty positive-cost intervals in that cohort remain open. The broad Borel upper-density theorem and the structural aggregation, reset, and fiber results retain their separate hypotheses and are preserved in the technical volumes.

The new evaluation is generated only after the final five numerical source files and protocol are frozen on the remote repository. Twelve primary environments vary state count, horizon, action count, discounting, allowance, initial occupancy, and tie geometry. Each method constructs its own enclosures and incumbent. Neither receives a saved policy, an exact operating value, or an earlier price. At the absolute target 10^{-3} , both formulations meet four of twelve primary targets, including one zero-cost case; two positive-cost closures require branching. On all eight unclosed cases the price formulation returns a smaller certified interval, but it produces no additional primary target hit in this sample. The separate observed-fleet problems have horizons two and three and positive costs. Both are certified under the registered target, with the price formulation’s widths below 10^{-8} . These results show a complete working pipeline and an exact continuum transfer in the stated class; they do not establish uniformly practical global optimization.

1.1. *Relation to constrained-control duality and approximation*

Occupation measures express conventional expected constrained Markov decision problems (Altman, 1999). Finite linear-programming approximations can converge, with quantitative rates under suitable assumptions (Dufour and Prieto-Rumeau, 2013, Saldi, 2018). Our all-restart constraint is different from an initial-distribution average of regrets. Theorem 1 derives the precise occupation program corresponding to the proposed prices, rather than inferring an equivalence from the common use of multipliers. Its common-budget completion is a finite nonconvex formulation of the original policy object. Theorem 2 shows that dropping this completion changes the value even with action-independent transitions.

Information-relaxation duality relaxes information restrictions and uses penalties to recover bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). The relaxation derived here instead separates incoming allocations of a shared successor regret. It does not relax nonanticipativity, and the policy remains Markov throughout. Constant prices reproduce an initial-regret Lagrangian; state-date prices can be strictly stronger, but their finite supremum is still the identified relaxation. This is a formal relation to established dual objects, not a claim to a new general principle of duality.

The lower-node constructions use classical McCormick inequalities (McCormick, 1976); the direct-value baseline uses aggregate continuation products. The enhanced construction uses signed witness-relative regret and disaggregated products with simplex identities. Both are read by the same independent arithmetic checker and searched by the same driver. This isolates a formulation comparison, not an externally authored solver competition. Verified residual correction follows the separation between numerical proposals and mathematical certificates emphasized by Rump (2010). The previously favorable native SCIP comparisons remain in the supplement with their uncertified lower bounds distinctly identified.

The continuum theorem below is not a generic state-quantization theorem. It is an exact forward-closure result, together with a globally feasible continuation extension. It complements rather than subsumes the approximation results of Dufour and Prieto-Rumeau (2013) and Saldi (2018). The economic example is a designed observed-fleet service contract. Unlike the estimated replacement application of Rust (1987), it supplies neither estimated primitives nor empirical implementation costs. High-dimensional operating approximation of the kind studied by Brumm and Scheidegger (2017) remains an important separate test; the current enclosures use finite directed Bellman passes.

2. THE COMMON-POLICY REVISION PROBLEM

Let X be a standard Borel space, $A = \{1, \dots, m\}$, and $t = 0, \dots, T - 1$. The known transition kernels P_t^a are Borel. Rewards, nonnegative implementation costs, and terminal rewards are Borel and satisfy

$$|r_t(x, a)| \leq R, \quad 0 \leq k_t(x, a) \leq K, \quad |g(x)| \leq G.$$

Throughout, $0 < \beta < 1$, $T < \infty$, and $\varepsilon > 0$. Define $S_h = \sum_{j=0}^{h-1} \beta^j$ and $S_0 = 0$. A policy p is a family of Borel probability vectors $p_t(\cdot | x)$. Its operating and implementation values and the optimal operating value satisfy

$$\begin{aligned} J_T^p &= V_T = g, & C_T^p &= 0, \\ V_t(x) &= \max_a \{r_t(x, a) + \beta P_t^a V_{t+1}(x)\}, \end{aligned} \tag{1}$$

$$J_t^p(x) = \sum_a p_t(a | x) \{r_t(x, a) + \beta P_t^a J_{t+1}^p(x)\}, \tag{2}$$

$$C_t^p(x) = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}.$$

For a specified initial probability law ν , the target is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \text{ for every } t, x\}. \tag{3}$$

An operating-optimal Borel policy exists by backward first-maximizer selection, so this feasible set is nonempty. We use an infimum when a general Borel optimizer has not been established. Finite models have a compact policy simplex and continuous constraints, hence attain the minimum.

Set

$$d_t(x, a) = V_t(x) - r_t(x, a) - \beta P_t^a V_{t+1}(x), \quad D_t^p = V_t - J_t^p. \quad (4)$$

Then $d_t \geq 0$, $D_T^p = 0$, and $D_t^p = \sum_a p_t(a | x)(d_t(x, a) + \beta P_t^a D_{t+1}^p(x))$. The economic restriction is $0 \leq D_t^p \leq \varepsilon$ everywhere. Its initial-law analogue, $\int D_0^p d\nu \leq \varepsilon$, is weaker and will be used only as a comparison.

For an installed rule a^0 , a representative cost is $k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}$. Physical switching from the previous action is a different friction and requires augmenting the state. A fixed lottery-administration fee f can be compared with a certificate in a designed budget exercise, but is not silently included in k . Scaling implementation costs by $c > 0$ scales K_R, L, U , and an appropriate absolute target by c , while leaving the operating constraint unchanged.

3. WHAT RESTART PRICES OPTIMIZE

For bounded nonnegative Borel prices λ_t with $\lambda_T = 0$, define $u_T = 0$ and

$$u_t(x) = \min_a \{k_t(x, a) + \lambda_t(x)(d_t(x, a) - \varepsilon) + \beta P_t^a[u_{t+1} + \varepsilon \min\{\lambda_t(x), \lambda_{t+1}(x)\}](x)\}. \quad (5)$$

Every feasible common policy satisfies $C_t^p + \lambda_t(D_t^p - \varepsilon) \geq u_t$. Indeed, backward substitution leaves the nonnegative hinge

$$(\lambda_t(x) - \lambda_{t+1}(y))_+ D_{t+1}^p(y) + (\lambda_{t+1}(y) - \lambda_t(x))_+ (\varepsilon - D_{t+1}^p(y)).$$

Together with $C_t^p \geq 0$, this proves the lower bound

$$L(\lambda) = \int \max\{0, u_0\} d\nu \leq K_R(\nu, \varepsilon). \quad (6)$$

The same induction permits lower disadvantages in place of d , an observation used by the implemented witness-only method.

3.1. An exact finite occupation interpretation

For this subsection let $X = \{1, \dots, n\}$. Write $P_{tia,j}$ for the transition probability and let μ be any nonnegative initial subprobability. For nonnegative variables y_{tia} define $z_{ti} = \sum_a y_{tia}$ and $f_{tij} = \beta \sum_a y_{tia} P_{tia,j}$. Edges only run from date t to date $t+1$, for $t < T-1$. Set missing edge sums to zero. Consider the linear program

$$\mathcal{O}(\mu) = \min_{y, \alpha, \gamma \geq 0} \sum_{t,i,a} k_{tia} y_{tia}, \quad (7)$$

$$z_{0i} = \mu_i, \quad z_{ti} = \sum_h f_{t-1,hi} \quad (t \geq 1),$$

$$\sum_a y_{tia}(\varepsilon - d_{tia}) \geq \sum_j \alpha_{tij} + \sum_h \gamma_{t-1,hi}, \quad (8)$$

$$\alpha_{tij} + \gamma_{tij} \geq \varepsilon f_{tij}. \quad (9)$$

The variables y are discounted occupations. The two nonnegative allocations on an incoming edge need not encode the same successor regret as allocations from other incoming states.

THEOREM 1—Price supremum and common-budget completion: *For every finite model satisfying Section 2,*

$$\sup_{\lambda \geq 0} \sum_i \mu_i u_{0i}(\lambda) = \mathcal{O}(\mu). \quad (10)$$

Both sides have finite attained optima. Adding variables $b_{ti} \in [0, \varepsilon]$, $1 \leq t < T$, and the equations

$$\alpha_{tij} = f_{tij} b_{t+1,j}, \quad \gamma_{tij} = f_{tij}(\varepsilon - b_{t+1,j}) \quad (11)$$

to (7)–(9) makes its optimum equal to the original all-restart common-policy value with objective measure μ . Thus price optimization solves an explicitly identified relaxation; restoring shared successor budgets solves the stated policy problem.

PROOF: Introduce free variables v_{ti} and nonnegative variables λ_{ti}, h_{tij} , with $v_T = \lambda_T = 0$. The price linear program maximizes $\sum_i \mu_i v_{0i}$ subject to

$$\begin{aligned} v_{ti} - \beta \sum_j P_{tia,j} v_{t+1,j} + (\varepsilon - d_{tia}) \lambda_{ti} - \beta \varepsilon \sum_j P_{tia,j} h_{tij} &\leq k_{tia}, \\ h_{tij} &\leq \lambda_{ti}, \quad h_{tij} \leq \lambda_{t+1,j}. \end{aligned} \quad (12)$$

There are no h variables in the last period. Given a price field, increasing h to the corresponding minimum can only relax these inequalities. Backward maximization of v then gives exactly (5). Conversely, (5) and those minima are feasible. This identifies the linear-program value with the supremum on the left of (10).

Assign dual multipliers y to (12) and α, γ to the two upper bounds on h . The free v variables give occupation conservation. The nonnegative λ variables give (8); the nonnegative h variables give (9). The resulting dual is precisely (7). It is feasible: for any feasible policy set $y_{tia} = \beta^t \Pr_\mu^p(X_t = i) p_t(a | i)$, $\alpha_{tij} = f_{tij} D_{t+1,j}^p$, and $\gamma_{tij} = f_{tij}(\varepsilon - D_{t+1,j}^p)$. At a positive-date node the budget inequality is an equality; at a root it is its all-restart regret inequality. The primal price program is also feasible, for example with zero prices and sufficiently small v . Its value is bounded by a feasible policy's cost. Finite-dimensional linear-program duality proves equality and attainment.

The same policy construction satisfies (11) with $b = D^p$, so the completed program has value at most the original optimum. In the reverse direction, take a completed feasible solution. At every node with $z_{ti} > 0$, put $p_t(a | i) = y_{tia}/z_{ti}$. Conservation ensures that a positive-probability transition of this policy goes to a positive-occupation node. For $t \geq 1$, substituting (11) in (8) and using incoming conservation gives

$$b_{ti} \geq \sum_a p_t(a | i) \{d_{tia} + \beta \sum_j P_{tia,j} b_{t+1,j}\},$$

with terminal budget zero. At roots of positive μ mass the same argument replaces b_{0i} by ε . Backward induction gives actual regret at most b at positive-date occupied nodes and at most ε at roots. At an unoccupied node choose a first maximizer of $r_t + \beta P_t J_{t+1}^p$ after the later policy has been completed. If the future policy is globally feasible, this yields $J_t^p \geq V_t - \beta \varepsilon \geq V_t - \varepsilon$. Occupied transitions do not enter unoccupied successors, so this extension does not change their values. It is globally feasible and has objective $\sum k y$. This proves the reverse inequality. *Q.E.D.*

The theorem does not say that a usual occupation-measure formulation is incapable of expressing common policies. It gives the additional bilinear compatibility which this particular price relaxation omits. The problem-specific content is the precise edge allocation and its completion, not linear-program duality itself.

COROLLARY 1—Clipping and constant multipliers: *In a finite model,*

$$\sup_{\lambda} L(\lambda) = \max_{S \subseteq \text{supp } \nu} \mathcal{O}(\nu \mathbf{1}_S). \quad (13)$$

For a constant price $\lambda_{ti} = \ell$ at all nonterminal dates, the un-clipped objective is

$$\sum_i \nu_i u_{0i} = \inf_p \left\{ \int C_0^p d\nu + \ell \left(\int D_0^p d\nu - \varepsilon \right) \right\}.$$

Consequently the best constant-multiplier bound is no larger than the un-clipped state–date–price supremum, which is no larger than K_R .

PROOF: For every vector v , $\sum_i \nu_i \max(0, v_i) = \max_S \sum_{i \in S} \nu_i v_i$. A finite maximum commutes with a supremum, so Theorem 1 proves (13). For the second statement, the price correction cancels successive constant-price allowance terms. Summing them along a trajectory leaves $-\ell\varepsilon$, while the disadvantages telescope to D_0^p . Backward minimization of the transformed objective proves the identity. *Q.E.D.*

The implementation uses a finite price cap and numerically proposed prices. It enumerates all nonempty masks when $n \leq 4$ and otherwise proposes the full mask and positive-mass singletons, subject to a declared time budget. Every proposed field is reconstructed rationally and ranked by its clipped certificate. A truncated mask search is not advertised as attaining (13); the earlier un-clipped-only objective is not silently equated with the displayed bound. Appendix A.2 gives an exact example in which the full-mask optimum is $3/8$ while a clipped field attains $15/32$.

3.2. A strict separation with exact rational values

THEOREM 2: *There is a two-state, two-period, two-action model with action-independent transitions for which the constant-multiplier optimum, the clipped restart-price supremum, and the common-policy optimum are respectively*

$$\frac{5}{4} < \frac{27}{16} < \frac{7}{4}. \quad (14)$$

The first two values have exact matching primal and dual witnesses. The final value follows from a one-variable affine minimization.

PROOF: Take $\beta = \varepsilon = 1/2$, zero terminal reward, and actions G, B with operating rewards $0, -1$. Every transition goes to state 1. At date 0 the cost of G is 1 in state 1 and 3 in state 2; at date 1 it is $3/2$ in either state. Action B costs zero. Let $\nu = (1/2, 1/2)$. Then $V = 0$ and $d(G) = 0, d(B) = 1$.

Let q denote the probability of B at date 1 in state 1 and q_i the two date-0 probabilities. Feasibility requires $0 \leq q \leq 1/2$ and $q_i \leq 1/2 - q/2$. Costs decrease with each q_i , so they equal these upper limits at an optimum. The objective becomes $7/4 + q/4$, minimized at $q = 0$ with value $7/4$.

Set $\lambda_0 = (1, 3)$ and $\lambda_1 = (3/2, 3/2)$. The recurrence gives $u_1 = (3/4, 3/4)$ and $u_0 = (9/8, 9/4)$, hence the lower value $27/16$. A feasible occupation witness in (7) takes the date-0 probabilities of B as $(1/4, 1/2)$ and the date-1 probability as $1/4$. The discounted occupations are $y_{01} = (3/8, 1/8)$, $y_{02} = (1/4, 1/4)$, and $y_{11} = (3/8, 1/8)$. On the two incoming edges use $(\alpha, \gamma) = (1/8, 0)$ and $(0, 1/8)$. Every occupation, node-budget, and edge inequality holds, and the objective is $27/16$. These matching witnesses prove the un-clipped supremum.

Clipping cannot improve it. With initial mass only at state 1 the original optimum is $9/8$ (choose $q = 1/2$); with mass only at state 2 it is $9/4$ (choose $q = 0$). Every price potential at the corresponding root is bounded by that positive optimum, hence so is its positive part. The same price field attains both bounds simultaneously.

Finally, for the expected-initial-regret problem choose $(q_1, q_2, q) = (0, 1, 0)$. Its average regret is $1/2$ and its cost is $5/4$. The constant price $\ell = 3$ gives $u_0 = (1/4, 9/4)$ and the same value. Weak duality proves the constant-multiplier optimum $5/4$. The conflicting preferred terminal randomizations explain the gap between the middle and final values: incoming edges cannot deploy different values of q at their common successor. *Q.E.D.*

The separation supplies a constructive reason to refine shared policy variables rather than assuming that more price proposals alone must close every interval. It also distinguishes a policy-to-certificate identity from a true optimization error. The former's clipping correction is explicitly negative: if its un-clipped nonnegative slack sum is $\mathcal{S}$, then $C^p(\nu) - L(\lambda) = \mathcal{S} - \int (-u_0)_+ d\nu$. We do not call this clipped expression a sum of nonnegative terms. The full earlier identity and its case-level diagnostics are retained in the supplement.

4. END-TO-END CERTIFICATION FROM OPERATING ENCLOSURES

The algorithm's inputs are the rational finite model, ε , an interval target $\eta > 0$, and a work budget. The only operating information supplied to price generation, candidate construction, or node relaxations is a pair of certified Bellman witnesses. Exact optimal operating values and inherited feasible policies are not inputs.

4.1. Enclosures and feasible policy generation

Let $\underline{V}, \bar{V}$ satisfy terminal bounds and the componentwise Bellman inequalities

$$\underline{V}_T \leq g \leq \bar{V}_T, \quad \underline{V}_t \leq \max_a (r_t^a + \beta P_t^a \underline{V}_{t+1}), \quad \bar{V}_t \geq \max_a (r_t^a + \beta P_t^a \bar{V}_{t+1}). \quad (15)$$

Backward induction gives $\underline{V} \leq V \leq \bar{V}$. Write $w_t = \|\bar{V}_t - \underline{V}_t\|_\infty$ and $\xi = \max_{t < T} w_t$. The implementation uses exact rational sums followed by outward dyadic rounding in each Bellman pass; it never computes an exact optimal V as an intermediate oracle. In representable tied examples, the two witnesses may coincide, which is disclosed rather than described as difficult approximation.

The two useful advantage quantities are

$$\underline{d}_{tia} = \max\{0, \underline{V}_{ti} - r_{tia} - \beta P_{tia} \bar{V}_{t+1}\} \leq d_{tia}, \quad (16)$$

$$d_{tia}^0 = \underline{V}_{ti} - r_{tia} - \beta P_{tia} \underline{V}_{t+1}. \quad (17)$$

The second need not be nonnegative. For $B_t^p = \underline{V}_t - J_t^p$, it gives the exact signed recursion $B_t^p = \sum_a p_{tia} (d_{tia}^0 + \beta P_{tia} B_{t+1}^p)$, with the appropriate actual terminal payoff. The lower-node program uses $\underline{V} - \varepsilon \leq J \leq \bar{V}$, not a fiction that $\underline{V}$ is an exact optimal value.

To create the initial incumbent, proceed backwards. Given the already constructed continuation, put

$$Q_a = r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x), \quad Z_a = k_t(x, a) + \beta P_t^a \hat{C}_{t+1}(x).$$

Minimize $\sum_a p_a Z_a$ over the probability simplex subject to $\sum_a p_a Q_a \geq \bar{V}_t(x) - \varepsilon$. An extreme minimizer has at most two positive actions and is found by enumerating feasible pure actions and pairs straddling the threshold. This is a conditional greedy construction, not a claim of globally optimal implementation. It supplies a feasible policy without any earlier run. A separately timed local nonlinear proposal can improve it; every resulting candidate is repaired and checked exactly.

LEMMA 1—Upper-witness repair: *Put $s = (1 - \beta)\varepsilon$ and assume $\xi < s$. If a common proposal q satisfies $J_t^q \geq V_t - \varepsilon - \delta$ everywhere, a backward repair produces $\hat{p}$ with $J_t^{\hat{p}} \geq \bar{V}_t - \varepsilon$ and $J_t^{\hat{p}} \geq J_t^q$. Its maximum row total-variation displacement, before deployment quantization, is at most*

$$\min \left\{ 1, \frac{\delta + \xi}{\delta + s} \right\}. \quad (18)$$

Rounding every action except an operating-best continuation action down to a mesh h and assigning the remaining probability to that best action preserves feasibility and adds at most $(m - 1)h$ to this bound.

PROOF: At a given step evaluate the original row of q against the repaired future. By induction its value v is no smaller than J_t^q . The best continuation action has value $M \geq V_t - \beta\varepsilon$. Its margin above $\bar{V}_t - \varepsilon$ is at least $s - \xi > 0$. If v is below the threshold, mix the original row with that best action using the minimal fraction. If the threshold deficit is z , then $z \leq \delta + \xi$ and the fraction is at most $z/(z + s - \xi) \leq (\delta + \xi)/(\delta + s)$. Otherwise no mixture is needed. Values weakly increase at each step. Downward rounding of nonbest probabilities moves at most $(m - 1)h$ probability mass to a weakly better action, so preserves both value inequalities. *Q.E.D.*

For policies whose rows differ by at most ρ in total variation, uniform perturbation constants sufficient here are

$$D_J = 2(RS_T + G)S_T, \quad D_C = K \sum_{t=0}^{T-1} \beta^t S_{T-t}. \quad (19)$$

Then $\|J^p - J^q\|_\infty \leq D_J \rho$ and $|C^p(\nu) - C^q(\nu)| \leq D_C \rho$. These conservative constants follow by bounding the range of each continuation action value and iterating the one-step perturbation inequality. They require no action separation.

4.2. Two node formulations and exact endpoints

A node is a box for all nTm probabilities, intersected rowwise with the simplex. At a nonempty box, rectangular Bellman minimization and maximization give exact bounds $J^{\min}, J^{\max}, C^{\min}, C^{\max}$ for every policy in the box. Each extremum is itself attained by a common Markov selector because the box is rectangular across state–date rows. A node can be eliminated if its simplex intersection is empty or if $J_{ti}^{\max} < \underline{V}_{-ti} - \varepsilon$ somewhere. These are rational conditions, not numerical infeasibility flags.

The direct-value formulation writes the original Bellman equations in J, C and uses McCormick envelopes for each aggregate product $p_{tia}(r_{tia} + \beta P_{tia} J_{t+1})$ and its implementation analogue. The price formulation writes the signed $B = \underline{V} - J$ equations and disaggregates products by successor. It imposes $\sum_a (p_{tia} B_{t+1,j}) = B_{t+1,j}$ and the analogous cost identity. It additionally uses the certificate planes

$$C_{ti} + \lambda_{ti}\{B_{ti} + w_{ti} - \varepsilon\} \geq u_{ti}, \quad C_{ti} \geq \max\{0, u_{ti}\}, \quad (20)$$

where $w_{ti} = \bar{V}_{ti} - \underline{V}_{ti}$ and u is reconstructed with (16). Since $D_{ti}^p \leq B_{ti}^p + w_{ti}$, the first plane follows from the price inequality for every truly feasible policy. The variable shift and product identities do not change the underlying common-policy object. Neither formulation dominates the other as a general polyhedral statement.

For a node relaxation $\min c'z$ subject to $Az = b$, $Ez \leq f$, and $l \leq z \leq u$, rounded proposals y and $\tau \geq 0$ give the exact rational lower endpoint

$$y'b - \tau'f + \sum_j \min\{r_j l_j, r_j u_j\}, \quad r = c - A'y + E'\tau. \quad (21)$$

No dual feasibility tolerance is needed: the residual is explicitly paid. The node lower bound is the maximum of this endpoint, zero, the rectangular cost bound, the reconstructed price bound, and its inherited parent bound. An exact common feasible policy supplies the upper endpoint. The global lower bound is the minimum over the complete leaf cover, clipped above by that feasible upper value. The checker reconstructs each row and split independently of the optimizer.

4.3. A matched finite stopping result

THEOREM 3—Witness-only finite completion: *Let a retained policy box have maximum row total-variation diameter at most ρ . Suppose (15) holds, $\xi < s$, and its rectangular operating test does not eliminate it. Evaluating and repairing any box member gives a cost at most*

$$\Omega(\rho, \xi, h) = D_C \left[\rho + \min \left\{ 1, \frac{D_J \rho + 2\xi}{D_J \rho + \xi + s} \right\} + (m-1)h \right] \quad (22)$$

above the box's rectangular cost lower bound. Consequently, choose witnesses and deployment mesh with

$$D_C \{\xi/s + (m-1)h\} \leq \eta/64, \quad \xi < s. \quad (23)$$

Without work caps, midpoint best-bound search that bisects a widest coordinate at least every fourth depth, and evaluates a box member, terminates with a valid interval of width at most η . The statement holds for both node formulations, even when every numerical LP proposal is unavailable.

PROOF: A retained box has $J^{\max} \geq \underline{V} - \varepsilon$ everywhere. Any member q differs from its rectangular operating-maximizing policy by at most ρ per row, hence

$$J^q \geq V - \varepsilon - (\xi + D_J \rho).$$

Apply Lemma 1 with $\delta = \xi + D_J \rho$. The rectangular cost minimizer is another policy in the same box, so its objective differs from $C^q(\nu)$ by at most $D_C \rho$. The repair and deployment change the cost by at most D_C times their maximum row displacement. This proves (22). Relaxation strengthening and inherited lower bounds can only improve the interval.

Under (23), the limit of (22) as $\rho \downarrow 0$ is at most $2D_C\xi/s + D_C(m-1)h \leq \eta/32$. Choose a positive coordinate width w_* such that (22), with $\rho = mw_*$, is below η . There are $d = nTm$ coordinates. Along a path, after at most $4d\lceil \log_2(1/w_*) \rceil$ levels every coordinate has width at most w_* : the scheduled widest-coordinate splits alone suffice. Thus every deeper retained node already has a repaired feasible policy within η of its lower bound. While the global interval exceeds η , best-bound search selects a leaf below the current upper value minus η , which cannot be such a sufficiently refined node. Only finitely many shallower nodes exist. Exact elimination cannot remove a feasible solution. The asserted stopping follows. *Q.E.D.*

The node bound implicit in this proof is exponential in the sufficient depth. It is an accuracy guarantee, not a polynomial complexity claim. What changes relative to an idealized exact-oracle theorem is that the implemented pipeline supplies the witnesses and the feasible upper policy appearing in the proof, and its arithmetic level depends on the requested target.

For lower-price conditioning, the implementation starts at 8 bits and adds 8 bits until (23) holds and

$$\Lambda \sum_{t=0}^{T-1} \beta^t (w_t + \beta w_{t+1}) \leq \eta/64, \quad \Lambda = 4096. \quad (24)$$

The cap is fixed before model generation. Since the price field is generated only afterwards, this is an *ex ante* conditioning rule, not a favorable exact-model price reused with noisy values. Directed finite Bellman widths tend to zero as precision increases, so the rule terminates. Exact rational residuals still certify any proposed multipliers, irrespective of their quality. There is no claim that increasing precision alone eliminates optimization slack.

5. A PAIRED CONTROLLED-CONTINUUM CONSTRUCTION

Consider the same Borel problem (3), with transition kernels

$$P_t^a(x, \cdot) = \sum_{s=1}^S \omega_{tas}(x) \delta_{F_{tas}(x)}(\cdot), \quad \omega_{tas} \geq 0, \quad \sum_s \omega_{tas} = 1, \quad (25)$$

where the maps and weights are Borel. They may be action dependent, noninvertible, and state dependent. Let the *objective* initial law have finite support H_0 , and recursively set

$$H_{t+1} = \bigcup_{x \in H_t, a, s: \omega_{tas}(x) > 0} \{F_{tas}(x)\}. \quad (26)$$

Equal successors are identified, not copied into history-specific nodes. This is essential: two paths that reach the same state and date must use the same future policy.

THEOREM 4—Finite observation and global extension: *Under Section 2 and (25), with a finitely supported objective initial law, the unrestricted Borel common-policy value equals the finite common-policy value on the forward sets (26):*

$$K_R^X(\nu, \varepsilon) = K_R^H(\nu, \varepsilon). \quad (27)$$

Every feasible finite policy has an explicit Borel extension satisfying the operating constraint at every point of X , not just at the forward sets. Therefore a paired finite certificate $[L, U]$ is a paired certificate for the full-state problem with exactly the same endpoints. The construction uses at most $|H_0|(mS)^t$ states at date t and incurs zero spatial approximation error.

PROOF: All possible successors of a point of H_t belong to H_{t+1} , so the operating Bellman value on these sets equals the restriction of V^X by backward induction. Every globally feasible policy restricts to a feasible finite policy and has the same objective. This proves $K_R^H \leq K_R^X$.

For the reverse direction take any finite feasible policy p^H . Put $\hat{J}_T = g$. After defining the later extension, set

$$\hat{p}_t(\cdot | x) = \begin{cases} p_t^H(\cdot | x), & x \in H_t, \\ \delta_{a_t(x)}, & x \notin H_t, \end{cases} \quad a_t(x) = \text{first maximizer of } r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x). \quad (28)$$

Finite sets are Borel. Finite-action first-maximizer selection and the Borel kernels make the extension Borel. Values on H_t are exactly the finite values because all successors stay on the forward sets. They satisfy $\hat{J}_t \geq V_t - \varepsilon$. Off H_t , assuming global feasibility at date $t + 1$, one has

$$\hat{J}_t(x) = \max_a \{r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x)\} \geq V_t(x) - \beta \varepsilon \geq V_t(x) - \varepsilon.$$

This completes the induction. Starting under ν , the policy never leaves the finite forward sets, so its implementation objective is unchanged. Taking infima proves the reverse inequality. Each state has at most mS successors, which gives the size bound. No state or probability is perturbed, establishing the zero spatial error. *Q.E.D.*

COROLLARY 2—Executable paired accuracy: *Suppose the finite forward model can be constructed with exact rational arithmetic, including decidable equality of successor states. Apply Theorem 3 to that model and use (28) off the forward sets. This gives a computable paired hierarchy and a stopping rule $U - L \leq \eta$ for (3). A running maximum of certified lower endpoints and running minimum of feasible upper endpoints give monotone sequences converging to the same value when the work caps are removed and targets tend to zero.*

The extension is also operational for the affine rational model below. At any queried state, evaluate its finitely branching continuation and test membership in the finite rational forward sets. This computes an action without an exact global operating oracle. In a piecewise-affine implementation, ordinary max-affine breakpoints coexist with exceptional preimages of the assigned finite policies. Both counts may grow exponentially with the horizon. The theorem guarantees a representation and a matched certificate, not cheap symbolic compilation. The publication contract checker independently reconstructs all forward states, merges, transition rows, rewards, and the terminal folding used in the experiment.

5.1. An observed-fleet service contract with original controlled atoms

The continuous condition is $x \in [0, 1]$. Operating rewards are $r(x, a) = x - c_a$, with

$$(c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2, \quad \beta = 19/20.$$

An installed action 0 costs nothing to retain; either departure costs $1 + x$. The two shock outcomes for the three actions are

$$\begin{aligned} F_0(x) &\in \{3x/4, 3x/4 + 1/20\}, \\ F_1(x) &\in \{17x/20 + 1/10, 17x/20 + 7/50\}, \\ F_2(x) &\in \{x/10 + 4/5, x/10 + 17/20\}, \end{aligned}$$

with probabilities $3/5, 2/5$. These are the original affine controlled atoms, not a finite-state replacement for different dynamics. Condition affects reward, revision expense, and every controlled successor. We fix $\varepsilon = 1/100$ and the observed initial fleet law $\nu = (3/4)\delta_{1/5} + (1/4)\delta_{4/5}$, at horizons 2 and 3.

A service organization can interpret the statewide allowance as a warranty on remaining operating value whenever its rule is restarted, including off-fleet states; the objective weights the vehicles in the observed fleet. This is a specified decision contract, not evidence that any particular organization uses it. The extension in Theorem 4 is what makes the warranty global despite the finite objective population. Lottery administration is assumed permitted by that contract and is charged separately if nonzero.

To illustrate decision units fixed before evaluation, assign \$500 to one normalized implementation unit and consider budget levels \$100, \$250, \$500, and \$1,000. The two certified optimal costs are approximately \$634.80 and \$1,434.35. Thus the two-period task has a feasible certified policy below \$1,000 but none below \$500; the three-period task cannot satisfy a \$1,000 implementation budget under the stated operating guarantee. These conclusions are robust to the reported optimization intervals. They are not estimates of market maintenance costs or empirical welfare. The \$500 scale and budget grid are explicit illustrative primitives, and a different scale changes the monetary conclusions mechanically.

For horizon 2 the active forward-set sizes are 2, 11, 63; for horizon 3 they are 2, 11, 63, 367. The last transition's expected g is folded into the last-period reward exactly, avoiding terminal padding. The policy problem has respectively 13 and 76 active decision nodes. Dummy nodes used for rectangular storage carry zero costs and have no incoming transitions from active states. Their equations are checked separately; they do not stand for omitted physical states. Table II reports the paired intervals.

5.2. What is not an initial-law approximation

Equation (27) keeps ν fixed. It does not authorize replacing a uniform initial law by finitely many atoms. In particular, the earlier thirty positive-cost uniform-initial-law intervals are unchanged, and the earlier nonlinear two-state widths remain roughly 25–76% at allowance $1/2$. Their original results and proofs are reproduced in the retained technical material. The one-sided Borel density theorem likewise remains a one-sided theorem outside the hypotheses of Theorem 4. A general matched numerical hierarchy for diffuse initial laws with controlled moving atoms requires further structure and is not inferred from finite observation.

6. FROZEN END-TO-END EVALUATION

6.1. Information, cases, and work accounting

The final numerical source and the model generator were frozen at commit `b636c40609863b498658db9d404676a5f74a236b`, before generating the twelve primary environments or the two fleet inputs. The development smoke test used a separate two-state model. The primary cases combine warranty maintenance, inventory, queueing, and time-varying ties, at three sizes:

$$(n, T, m) \in \{(3, 4, 3), (5, 8, 3), (8, 12, 4)\},$$

$$\beta \in \{7/8, 15/16, 63/64\}, \quad \varepsilon \in \{1/50, 1/100, 1/200\}.$$

The parameters vary jointly; these rows do not identify a separate causal effect of each dimension. Nonuniform initial probabilities are used, with the largest level placing mass only on two states.

TABLE I
ALL TWELVE SOURCE-FROZEN PRIMARY CASES

Case	Price upper	Direct width	Price width	Price rel.%	P nodes	P stop
W0	3.110700	0.000995	0.000987	0.0317	5	target
W1	8.236030	0.128	0.0126	0.1535	169	time_cap
W2	8.926491	0.0447	0.0111	0.1243	27	time_cap
I0	1.982232	0.000997	0.000892	0.0450	9	target
I1	5.044754	0.0509	0.0123	0.2440	203	time_cap
I2	10.244317	0.0718	0.0109	0.1061	23	time_cap
Q0	0.230546	9.3e-09	9.3e-09	0.0000	1	target
Q1	7.424602	0.0238	0.00237	0.0320	191	time_cap
Q2	10.652032	0.0506	0.0117	0.1102	29	time_cap
T0	1.074850	0.151	0.00855	0.7953	255	node_cap
T1	0.000000	0	0	0.0000	1	target
T2	6.135079	0.95	0.0524	0.8537	39	time_cap

W, I, Q, and T denote warranty, inventory, queue, and ties. Indices 0–2 have the sizes listed in the text. The primary criterion is absolute width $\leq 10^{-3}$; *target* denotes only that criterion. Relative width divides by the upper endpoint (zero is displayed for the exact zero-cost row). Direct and price upper endpoints, all costs, and all trajectories appear separately in the supplement. Rounded table entries do not replace the exact endpoints used to decide success.

The tie family has nonzero, time-varying operating values and different optimal faces across states and dates; it is one generator family, not three independent applications. Its hidden construction potential is not passed to either solver. The three other families have genuinely nonzero witness widths. No primary outcome was used to alter the frozen solver.

Each method starts from the same raw model and independently constructs its own witnesses and initial policy. A conditional greedy policy and a local proposal are charged to that run. Prices are generated from the witnesses, never from exact-run prices. The methods have the same 30-second soft construction allocation, 255-node cap, 3-second local allocation, and a 3-second allocation for each LP proposal. The price allocation is capped at 5 seconds *within* construction. Exact certificate operations finish before a budget check. Reported all-in time additionally includes encoding, compression, and independent replay, and is not capped at 30 seconds. Thus the comparison has equal declared construction rules and measured cumulative costs, not an assertion that every run consumes exactly the same total CPU time.

The environment is Python 3.13 with NumPy 2.3.5 and SciPy 1.17.0, one numerical thread, on GitHub-hosted Linux runners. Family jobs use separate machines; direct and price runs are sequential within a family process. Process high-water memory is therefore not a method-isolated peak. The supplement retains that limitation and reports all original timings. Floating optimization is only a proposal mechanism; both lower endpoints and deployed policies have exact rational checks.

6.2. Primary results: tighter intervals but unchanged target counts

Table I reports the complete primary sample. Both methods have four target hits: W0, I0, Q0, and T1. T1 is zero cost, Q0 closes at the root, and W0 and I0 require branching. The price formulation needs respectively 5 and 9 nodes on the branching successes, compared with 69 and 63 for the direct formulation. All four successes are new inputs generated after the freeze, but the price formulation adds *zero* method-relative target hits on this sample. This is different from counting cumulative historical closures.

TABLE II

PAIRED CERTIFICATES FOR THE ORIGINAL AFFINE ATOMS WITH AN OBSERVED INITIAL FLEET

Case	Price upper	Direct width	Price width	Price rel.%	P nodes	P stop
F0	1.269602	2.28e-09	2.28e-09	0.0000	1	target
F1	2.868705	0.000367	8.27e-09	0.0000	1	target

F0 and F1 have horizons 2 and 3. Their common initial law is $(3/4)\delta_{1/5} + (1/4)\delta_{4/5}$. Theorem 4, not a state-quantization approximation, makes these intervals valid for the full Borel-state contract. Both methods generate their own incumbents. These are two designed contracts, not independent empirical applications or closures of uniform-initial-law rows.

All eight cases left open by the direct formulation remain open with prices, but each has a smaller price interval. The largest price widths are about 0.01110, 0.01086, and 0.01174 in W2, I2, and Q2. T2 still has width 0.05238, although below the direct width 0.95040. These are informative improvements, not successful applications of the 10^{-3} stopping rule. T0 also reaches the node cap with a nonzero width. Increasing precision cannot remove these optimization gaps: the tied witnesses already have zero width in exact dyadic arithmetic.

Across the twelve primary cases, the direct method uses about 286.26 seconds of measured all-in work and the price method about 340.52 seconds. The largest individual all-in time is about 53.42 seconds. The more expensive price relaxation trades fewer processed nodes for tighter lower values; it is not declared a uniform speed winner. Full per-stage costs and proof sizes are available in Supplement Tables S2–S4, while every recorded lower/upper improvement appears in the machine-readable trajectory file. The search refinement and feasible upper generation are actually executed; these rows are not a price-only attachment to old incumbents.

6.3. Observed-fleet intervals and the full-state interpretation

The price widths are approximately 2.28×10^{-9} and 8.27×10^{-9} ; the direct three-period width is about 3.67×10^{-4} . All four intervals meet the predeclared target. Their independent arithmetic checks concern the finite forward model, while the independently written contract check reconstructs the exact original affine maps and terminal folding. The extension proof transfers those checks to the unrestricted Borel-policy target. It is an analytic theorem, not a machine-checked proof of global Borel measurability. The zero spatial-error assertion follows from exact forward closure and should not be attributed to the finite reader alone.

6.4. Precision and verification boundaries

The adaptive rule chooses 32, 40, and 48 bits across the three non-tie size levels and 32 bits for the fleet cases. The tied construction is exactly representable at the starting 8-bit witness level; deployment still uses at least 32 bits. Every table reports price magnitude, the actual witness guard ratio, and the price-amplification budget. Large prices and high discounting impose a real enclosure requirement; the reported finite passes do not demonstrate a cheap high-dimensional learned or sparse-grid oracle.

All 28 proof objects are independently reconstructed from raw primitives, witnesses, prices, complete covers, and policies. The publication stage repeats this reconstruction without re-running the optimizers and adds model-contract and corruption checks. This separates source identity, finite arithmetic validity, analytic validity, and performance generalization. The two formulations are separately reconstructed but share a search driver and proposal routine; they are not independent external replications. The historical SCIP native lower bounds remain outside the rational portfolio and are preserved as numerical comparisons, not relabeled certified results.

6.4.0.1. *Post-execution diagnostics.* A separate diagnostic program computes exact finite operating values only after all frozen prices, policies, and proof objects have been produced. It checks the policy-to-certificate identity and the fixed-field witness-loss inequality on the fourteen price runs. These exact values are not inputs to any price proposal, policy proposal, repair, branching decision, or reported endpoint. The supplement reports the clipping correction separately from the three nonnegative slacks, and separates price-certificate width from the final tree width.

7. IMPLICATIONS AND REMAINING NUMERICAL QUESTIONS

The results identify the common continuation as the organizing constraint. Restart prices have a precise occupation-flow meaning; the missing shared budgets can be written explicitly and restored by a convergent finite refinement. A finite solver can construct both endpoints using only certified operating enclosures. For finitely observed initial populations and controlled finite-atomic dynamics, exact forward closure and global extension turn those finite intervals into paired full-state certificates.

The successful fleet calculations and the new source-frozen sample answer different questions. The former exercise a matched continuous-state theorem on original action-dependent atoms with an observed initial law. The latter show that an end-to-end witness-only implementation works without privileged exact-model discoveries, while exposing remaining failures under finite budgets. Four of twelve primary targets are met by each formulation, and eight remain open. These results do not imply polynomial search complexity, superiority to modern uncertified global solvers, or an empirical welfare conclusion.

The diffuse-initial-law continuous problem remains consequential. Its old intervals are retained without revision of their endpoints; finite observations are not used as a substitute initial law in those calculations. Broader settings involving continuous actions, unbounded primitives, learned kernels, or multiple operating constraints also require their own approximation and feasibility arguments. The inherited lower-only multicriterion extension is not promoted into an unproved general repair theorem. The present contribution keeps the original common-policy target and supplies an explicit algorithmic chain where the stated hypotheses make that chain complete.

APPENDIX A: ADDITIONAL ERROR AND DUAL DETAILS

A.1. Operating-witness sensitivity of a price field

Suppose u is reconstructed with exact disadvantage d , and $\underline{u}$ with (16), using the same nonnegative field. The inequalities

$$0 \leq d_{tia} - \underline{d}_{tia} \leq w_t + \beta w_{t+1}$$

follow directly from the operating enclosure (including the clipping at zero). Backward minimum stability yields

$$0 \leq u_{ti} - \underline{u}_{ti} \leq \Gamma_t, \quad \Gamma_t = \sum_{s=t}^{T-1} \beta^{s-t} \|\lambda_s\|_{\infty} (w_s + \beta w_{s+1}).$$

The positive-part map is 1-Lipschitz, so the same Γ_0 bounds the change in the displayed clipped lower endpoint. Any directed approximation of the price expectation adds its certified one-step error to this recursion. In the executed model the expectation is an exact rational finite sum. This is a perturbation result for a *fixed* field; the algorithm's cap-based rule is imposed before generating the field, so it does not presuppose stability of an exact-model price optimizer.

A.2. Clipping can change the optimizing price

The mask maximization is not merely a different display of the full-mask optimum. Modify the two-period example in Theorem 2 as follows. Keep $\beta = \varepsilon = 1/2$ and deterministic transitions to state 1. At date 0, rewards of G, B are 0, -1 , while costs are $(0, 0)$ in state 1 and $(3, 0)$ in state 2. At date 1, rewards of G, B are 0, $-1/4$ and costs are $(1, 0)$ in both states. Let $\nu = (3/4, 1/4)$ and $g = 0$. The operating value is zero.

Let l be the price at the reached terminal decision state. Its potential is

$$u_1(l) = \min\{1 - l/2, -l/4\}.$$

For this fixed l , the best root potentials use prices 0 and 3 and equal $u_{01} = u_1(l)/2$ and $u_{02} = 3/2 + \min(3, l)/4 + u_1(l)/2$. The first claim follows from the zero-cost good action; the second follows by checking the positive slope before root price 3 and the negative slope thereafter. Thus the optimized full-mask objective is

$$3/8 + \min(3, l)/16 + u_1(l)/2,$$

whose maximum $3/8$ occurs at $l = 0$. Its clipped value there is also $3/8$. But $l = 3$ gives potentials $(-3/8, 15/8)$ and clipped value $15/32 > 3/8$. The latter is the actual common-policy optimum: take the terminal bad action with probability 1, use the root bad action with probability $3/8$ in state 2, and use the good action in state 1. Its operating shortfalls obey all restart bounds, its objective is $15/32$, and the displayed certificate matches it. Hence optimizing the un-clipped objective can select a strictly inferior clipped certificate. Enumerating the relevant initial masks fixes this mismatch without altering the validity of any field.

A.3. A quantitative form of repaired Borel upper density

The inherited upper-density result can be stated with a finite error modulus, although not with a universal computable partition-size rate. Fix a globally feasible policy p and define finite dominating measures

$$\mu_0 = \nu, \quad \mu_{t+1} = \sum_{a=1}^m \mu_t P_t^a.$$

Their total masses are m^t . For any Borel proposal q , let $e_t(x) = \text{TV}(q_t(\cdot | x), p_t(\cdot | x))$ and $E_t = \int e_t d\mu_t$. Put $M = 2(RS_T + G)$, $s = (1 - \beta)\varepsilon$, $c = 1 + M/s$, and

$$A_t = cE_t + \frac{Mc\beta}{s} \sum_{v=t+1}^{T-1} (\beta c)^{v-t-1} E_v,$$

$$\mathcal{B}(E) = K \sum_{t=0}^{T-1} \beta^t S_{T-t} A_t.$$

Then exact continuation-best repair of q gives a globally feasible $\hat{p}$ with

$$|C_0^{\hat{p}}(\nu) - C_0^p(\nu)| \leq \mathcal{B}(E). \quad (29)$$

To verify this, write $a_t = \text{TV}(\hat{p}_t, p_t)$ and $\Delta_t = |J_t^{\hat{p}} - J_t^p|$. The threshold deficit before repair is at most $Me_t + \beta P_t^p \Delta_{t+1}$ because p is feasible. The best repaired continuation has margin at

least s , so the repair mass is at most this deficit divided by s . Therefore

$$a_t \leq ce_t + (\beta/s)P_t^p \Delta_{t+1}, \quad \Delta_t \leq Mce_t + \beta cP_t^p \Delta_{t+1}.$$

Integrating and using $\mu_t P_t^p \leq \mu_{t+1}$ proves $\int a_t d\mu_t \leq A_t$. The implementation performance-difference expansion and the range bound KS_{T-t} then give (29).

Rational simple proposals can approximate p in these finitely many $L_1(\mu_t)$ norms; the displayed expression gives value convergence after repair. But the dominating masses and factors c can grow rapidly, arbitrary Borel integrals need not be effectively evaluable, and a repaired simple proposal need not itself remain simple. Thus (29) is not a two-sided computable scheme for a diffuse initial law. The finite-observation theorem in the main text obtains such a scheme by exact closure, not by suppressing these difficulties.

A.4. Exact finite products and common budgets

For $p \in [l, u]$, $v \in [a, b]$, each product $z = pv$ obeys the sign-safe inequalities

$$\begin{aligned} z &\geq ap + lv - la, & z &\geq bp + uv - ub, \\ z &\leq ap + uv - ua, & z &\leq bp + lv - lb. \end{aligned}$$

The displayed four McCormick inequalities, not any sign shortcut, are reconstructed by the reader. Signed $B = \underline{V} - J$ variables are allowed. For a probability simplex, $\sum_a p_a v = v$ supplies an exact product conservation identity. These constraints preserve one continuation variable per successor. The occupation completion (11) is an alternative exact formulation, while the implemented probability-box search enforces the same compatibility through the original Bellman equations. The occupancy LP itself is not falsely described as complete for K_R .

APPENDIX B: ORGANIZATION OF RETAINED MATERIAL

The numerical supplement contains all 28 new rows, all recorded search trajectories, oracle conditioning, work components, the complete new source/test contract, and a compact assumption matrix. It also reproduces the complete R47 main article and supplement, which retain the strict-gap quadratic analysis, earlier finite searches, exact aggregation, reset transfer, observable fibers, original uniform-initial-law certificates, and nonlinear examples. A separate historical volume preserves the earlier R47 historical material. No earlier source, result, proof object, or review is deleted or modified. The preservation manifest checks that statement against the immutable reviewed baseline; preservation itself is not evidence for a new theorem or numerical target hit.

REFERENCES

- Altman, E. (1999): *Constrained Markov Decision Processes*. Chapman and Hall/CRC. [2]
- Brown, D. B., and J. E. Smith (2014): “Information Relaxations, Duality, and Convex Stochastic Dynamic Programs,” *Operations Research*, 62(6), 1394–1415. doi:10.1287/opre.2014.1322. [3]
- Brown, D. B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, part 1), 785–801. doi:10.1287/opre.1090.0796. [3]
- Brumm, J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85(5), 1575–1612. [3]
- Dufour, F., and T. Prieto-Rumeau (2013): “Finite Linear Programming Approximations of Constrained Discounted Markov Decision Processes,” *SIAM Journal on Control and Optimization*, 51(2), 1298–1324. doi:10.1137/120867925. [2, 3]

- McCormick, G. P. (1976): “Computability of Global Solutions to Factorable Nonconvex Programs: Part I—Convex Underestimating Problems,” *Mathematical Programming*, 10, 147–175. doi:10.1007/BF01580665. [3]
- Rump, S. M. (2010): “Verification Methods: Rigorous Results Using Floating-Point Arithmetic,” *Acta Numerica*, 19, 287–449. [3]
- Rust, J. (1987): “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher,” *Econometrica*, 55, 999–1033. doi:10.2307/1911259. [3]
- Saldi, N. (2018): “Finite-State Approximations to Discounted and Average Cost Constrained Markov Decision Processes,” arXiv:1807.02994. [2, 3]